1. Which of the following statements is correct?
   A. We can assume a "spherical cow" when analyzing the motion of a cow's leg.
   B. We can safely assume a cow is a point mass when analyzing its position change in the scale of a mile over a certain time period.
   C. We can never assume the cow is in a vacuum, even when we need to ignore the effect of air resistance.
   D. None of the above is correct.

2. Which of the following equations is correct?
   A. 1 ¢ = 1 $
   B. 1 = 0.01
   C. 1 ¢ = 0.01 $
   D. 100 ¢ = 0.01 $

3. Which option has the incorrect unit?
   A. Energy: joule
   B. Velocity: meter/second
   C. Acceleration: meter^2/second^2
   D. Force: kilogram · meter/second^2

4. The raven is sent from the Castle Black, which is at the origin, observing the white walkers at the coordinates $\vec{r}_4 = 3\hat{\imath} + 6\hat{\jmath}$ at the beginning of season 4, where length units are in kilofurlongs. Which of the following statements is correct?
   A. The distance between the white walkers and the Castle Black is 9 kilofurlongs.
   B. The distance between the white walkers and the Castle Black is 6 kilofurlongs.
   C. The distance between the white walkers and the Castle Black is 3 kilofurlongs.
   D. None of the above is correct.

5. If the displacement of an object is given by the function $\vec{s} = ((1.3m/s^3)t^3 + (2.7m/s^2)t^2)\hat{\imath}$, which of the following statement is correct?
   A. This is a constant acceleration motion.
   B. The magnitude of the acceleration is 2.7 m/s^2
   C. The magnitude of the acceleration at t = 0 is 0 m/s^2
   D. The magnitude of the acceleration at t = 0 is 5.4 m/s^2

6. An object takes 3 second to move from position A ($\vec{r}_A = -4.0\hat{\imath} + 10\hat{\jmath}$) to position B ($\vec{r}_B = 2.0\hat{\imath} + 5.0\hat{\jmath}$) along a straight path, where length units are in meters. Which of the following statements is correct?

A. The moving distance is 7.8 m.
B. The speed at t = 0 must be 2.6 m/s.
C. The acceleration at t = 0 must be 0.87 m/s^2
D. None of the above is correct.

7. A Chinook salmon can jump out of water with a speed v_0 and an angle θ above the horizontal. What is the speed of the fish when it is at the maximum height?
   A. v_0.
   B. $v_0 \cos\theta$.
   C. $v_0 \sin\theta$.
   D. 0.

8. A spaceship is traveling with an initial velocity of $\vec{v}_0 = (20.1\ m/s)\hat{\imath}$ when its rockets fire, imparting an acceleration of $\vec{a} = (2.02\ m/s^2)\hat{\imath} + (4.03\ m/s^2)\hat{\jmath}$. Based on this scenario, which of the following statements is correct?
   A. The x-component of the spaceship's velocity remains constant.
   B. The y-component of the spaceship's velocity remains constant.
   C. Both the direction of motion and speed change.
   D. None of the above is correct.

9. If the magnitudes of both charges are tripled and their separation is also tripled, what happens to the magnitude of the electrostatic force that each charge exerts on the other?
   A. It stays the same.
   B. It decreases by a factor of $3^{1/2}$.
   C. It increases by a factor of 3.
   D. It increases by a factor of 9.
   E. It increases by a factor of $3^{1/2}$.

10. An object can escape a planet with radius *R* and mass *M* at an escape speed given by $v = \sqrt{2GM/R}$, where G is the universal gravitational constant. Based on this expression for escape speed, which of the following statements is correct?
    A. A heavier object requires a higher speed to escape from the planet.
    B. A lighter object requires a lower speed to escape from the planet.
    C. An object requires a higher speed to escape from a planet with smaller mass and a larger radius.
    D. An object requires a higher speed to escape from a planet with greater mass and a smaller radius.

11. The force $\vec{F} = (3.04\text{N})\hat{\imath} + (4.01\text{N})\hat{\jmath}$. acts on a particle as it moves along the x axis from x = 0.33 m to x = 5.03 m. What is the work, in joules, done by the force on the particle?
    A. 20.5 J
    B. 14.3 J
    C. 5.00 J
    D. 35.7 J

12. A 4.5 kg cart initially moving at 2.1 m/s collides with a wall and rebounds at a speed of 1.3 m/s moving backward. The collision lasts for 0.2 seconds. What is the average force exerted on the cart during the collision?
    A. 47.3 N.
    B. 18 N.
    C. 76.5 N.
    D. None of the above is correct.

13. A 78kg hockey goalie is originally at rest protecting his goal. The opposing team attempts a shot on goal, however the goalie catches the 170g puck with his glove. The initial velocity of the puck before the goalie catches it is 150 km/hr. What is the speed of the goalie after catching the puck?
    A. 0.077 m/s
    B. 0.091 m/s
    C. 102 m/s
    D. 28.6 m/s